MODURIDE TABLET

DESCRIPTION: A 7 mm round, scored tablet, light purple in colour with markings 'd' and 'DUO 861'.

COMPOSITION: Each tablet contains Amiloride HCl 5 mg, Hydrochlorothiazide 50 mg.

PHARMACODYNAMICS: Clinical Pharmacology: Moduride provides diuretic and antihypertensive activity (principally due to the hydrochlorothiazide component), while acting through the amiloride component to prevent the excessive potassium loss that may occur in patients receiving a thiazide diuretic. Due to its amiloride component, the urinary excretion of magnesium is less with Moduride than with a thiazide or loop diuretic used alone. The onset of the diuretic action of Moduride is within 1 to 2 hours and this action appears to be sustained for approximately 24 hours.

Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High HCTZ use (≥50,000mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.68-4.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to HCTZ: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use (~25,000mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose (~100,000mg).

PHARMACOKINETICS: Amiloride HCl: Amiloride HCl usually begins to act within two hours after an oral dose. Its effect on electrolyte excretion reaches a peak between 6 and 10 hours and lasts about 24 hours. Peak plasma levels are obtained in 3 to 4 hours and the plasma half-life varies from 6 to 9 hours. Effects on electrolytes increase with single doses of amiloride HCl up to approximately 15 mg.
Amiloride HCl is not metabolized by the liver but is excreted unchanged by the kidneys. About 50 percent of a 20 mg dose of amiloride HCl is excreted in the urine and 40 percent in the stool within 72 hours. Amiloride HCl has little effect on glomerular filtration rate or renal blood flow. Because amiloride HCl is not metabolized by the liver, drug accumulation is not anticipated in patients with hepatic dysfunction, but accumulation can occur if the hepatorenal syndrome develops.
Hydrochlorothiazide: The mechanism of the antihypertensive effect of thiazides is unknown. Hydrochlorothiazide does not usually affect normal blood pressure.
Hydrochlorothiazide is a diuretic and antihypertensive agent. It affects the renal tubular mechanism of electrolyte reabsorption. Hydrochlorothiazide increases excretion of sodium and chloride in approximately equivalent amounts. Natriuresis may be accompanied by some loss of potassium and bicarbonate.
After oral use diuresis begins within two hours, peaks in about four hours and lasts about 6 to 12 hours.
Hydrochlorothiazide is not metabolized but is eliminated rapidly by the kidney. When plasma levels have been followed for at least 24 hours, the plasma half-life has been observed to vary between 5.6 and 14.8 hours. At least 61 percent of the oral dose is eliminated unchanged within 24 hours. Hydrochlorothiazide crosses the placenta but not the blood-brain barrier and is excreted in breast milk.

INDICATIONS: Moduride is indicated in the treatment of patients with oedema of cardiac origin, in hepatic cirrhosis with ascites and in patients with hypertension in whom potassium depletion might be anticipated. Moduride with its combination of Amiloride HCl and Hydrochlorothiazide, minimizes the possibility of the development of excessive potassium loss in patients during vigorous diuresis for prolonged periods. Moduride with its built in potassium sparing agent, is especially indicated in those conditions where the positive effect on potassium balance is particularly important.
Moduride may be used alone, or as an adjunct to other anti-hypertensive drugs. Since it enhances the action of these agents, the dosage of these antihypertensive drugs may need to be reduced to avoid the risk of an excessive drop in pressure.
Oedema of Cardiac origin: In these patients Moduride may provide increased sodium, chloride and water excretion and decreased potassium excretion. The positive effect of Moduride on potassium balance may be especially important for digitalized cardiac patients in whom potassium depletion can precipitate digitalis intoxication with potentially serious cardiac arrhythmias.

RECOMMENDED DOSAGE: As with all diuretic therapy, the rate of weight loss and serum electrolyte levels should determine the dose. After initiating diuresis the most satisfactory rate of weight loss is generally about 0.5 to 1.0 kg daily.
Oedema of cardiac origin: Moduride may be started at a dosage of 1 or 2 tablets a day. Dosage may be increased if necessary, but must not exceed 4 tablets a day. The optimal dosage is determined by the diuretic response and the serum potassium level. Once an initial diuresis has been achieved, reduction in dosage should be attempted for maintenance therapy. Maintenance therapy may be on an intermittent basis.
Hypertension: The usual dosage is one or two tablets given once a day or in divided doses. The dosage may be increased if necessary, but must not exceed four tablets a day.
Since hydrochlorothiazide may potentiate the action of other antihypertensive agents, the dosage of these agents may need to be reduced prior to the addition of Moduride to the regimen. This applies particularly to the ganglion blocking agents where a reduction of 50% at least is recommended.
Hepatic cirrhosis with ascites: Treatment should be initiated with a small dose of Moduride (1 tablet once a day). If necessary, dosage may be increased gradually until there is effective diuresis. The dosage should not exceed four tablets per day.
Maintenance doses may be lower than those required to initiate diuresis; therefore, reduction in the daily dose should be attempted when the patient's weight is stabilized. Gradual weight reduction in cirrhotic patients is especially desirable to reduce the likelihood of untoward reactions associated with diuretic therapy.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS: Hyperkalaemia: Moduride should not be used in the presence of elevated plasma potassium levels (interpreted as over 5.5 mmol per litre).
Antikaliuretic therapy or potassium supplementation: Other antikaliuretic agents and potassium supplements or a potassium-rich diet are contraindicated in patients receiving Moduride (such combination therapy is commonly associated with rapid increased in plasma potassium levels).
Impaired renal function: Anuria, acute renal failure, severe progressive renal disease and diabetic nephropathy are contraindications to the use of Moduride. Patients with increases in blood urea nitrogen (BUN) over 10.7 mmol / L, in serum creatinine levels over 0.13 mmol / L, or in whole blood urea values over 10.0 mmol / L, should not receive the drug without careful, frequent monitoring of serum electrolytes in BUN levels. Potassium retention in the presence of renal impairment is accentuated by the addition of an antikaliuretic agent and may result in the rapid development of hyperkalaemia.
In children: The safety for use of Amiloride HCl in children has not been established; therefore, Moduride is not recommended in the paediatric age group.
Known sensitivity to the drug: Moduride is contraindicated in patients who are hypersensitive to this product, or to other sulphonamide-derived drugs.

WARNING & PRECAUTIONS: Diabetes mellitus: In diabetic patients, hyperkalaemia has commonly occurred during therapy with Amiloride HCl, particularly if chronic renal disease or pre-renal azotaemia is present. Therefore, before initiating therapy in diabetic or suspected diabetic patients, the status of renal function should be known.
Insulin requirements in diabetic patients may be increased, decreased, or unchanged due to the Hydrochlorothiazide component. Diabetes mellitus which has been latent may become manifest during thiazide administration.
Metabolic or respiratory acidosis: Antikaliuretic therapy should be instituted only with caution in severely ill patients in whom respiratory or metabolic acidosis may occur, such as patients with cardiopulmonary disease and patients with decompensated diabetes. Shifts in acid-base balance alter the balance of extracellular / intracellular potassium and the development of acidosis may be associated with rapid increases in serum potassium levels.
Hyperkalaemia: Hyperkalaemia, defined as serum potassium levels over 5.5 mmol per litre, has been observed in patients who received Amiloride HCl, either alone or in combination with other diuretic drugs. This has been noted particularly in aged patients, diabetic patients and in hospitalized patients with hepatic cirrhosis or cardiac oedema who have known renal involvement, are seriously ill, or are undergoing vigorous diuretic therapy. These patients should be monitored carefully for clinical, laboratory and electrocardiographic (ECG) evidence of hyperkalaemia. Some deaths have been reported in this group of patients. Warning signs of hyperkalaemia include paraesthesia, muscular weakness, fatigue, flaccid paralysis of the extremities, bradycardia, shock and serum potassium and ECG abnormalities. Careful monitoring of the plasma potassium level is important because hyperkalaemia is not always associated with an abnormal electrocardiogram. When abnormal, the electrocardiogram hyperkalaemia is characterized primarily by tall, peaked T-waves or elevations from previous tracings. There may also be lowering of the R-wave and increased depth of the S-wave, widening and even disappearance of the P-wave, progressive widening of the QRS complex and prolongation of the PR interval and ST depression.
Treatment of hyperkalaemia: Should hyperkalaemia occur in patients taking Moduride, the drug should be discontinued immediately, and if necessary, active measures taken to reduce the plasma potassium level. Discontinuation of antikaliuretic therapy should be followed by intravenous administration of molar sodium lactate or oral or parenteral glucose with a rapid-acting insulin. If needed, a cation exchange resin such as sodium polystyrene sulphonate may be given orally or by enema. Patients with persistent hyperkalaemia may require dialysis.
Electrolyte imbalance and Reversible BUN increases: Determination of serum electrolytes to detect possible electrolyte imbalance should be performed at appropriate intervals.
Patient should be observed for clinical signs of fluid or electrolyte imbalance: ie. hyponatraemia, hypochloraemic alkalosis, hypokalaemia and hypomagnesaemia. It is particularly important to make serum and urine electrolyte determinations when the patient is vomiting excessively or receiving parenteral fluids. Warnings signs or symptoms of fluid and electrolyte imbalance include: dryness of mouth, thirst, weakness, lethargy, drowsiness, restlessness, seizures, confusion, muscle pains or cramps, muscular fatigue, hypotension, oliguria, tachycardia, and gastrointestinal disturbances such as nausea and vomiting. Hyponatraemia and hypochloraemia may occur during the use of thiazide and other oral diuretics even when amiloride HCl is used. Any chloride deficit during thiazide therapy is generally mild and may be lessened by the amiloride HCl component of Moduride. Hypochloraemia usually does not require specific treatment except under extraordinary circumstances (as in liver disease or renal disease). Dilutional hyponatraemia may occur in oedematous patients in hot weather; appropriate therapy is water restriction, rather than administration of salt, except in rare instances when the hyponatraemia is life-threatening. In actual salt depletion, appropriate replacement is the therapy of choice. Hypokalaemia may develop during thiazide therapy especially with brisk diuresis, when severe cirrhosis is present, during concomitant use of corticosteroids or ACTH, or after prolonged therapy. However, this is usually prevented by the amiloride HCl component of Moduride. Interference with adequate oral electrolyte intake will also contribute to hypokalaemia. Hypokalaemia may cause cardiac arrythmia and may sensitize or exaggerate the response of the heart to the toxic effects of digitalis (eg. increased ventricular irritability). Reversible increases in BUN levels have been reported; these have accompanied vigorous fluid elimination, especially when diuretic combinations were used in seriously ill patients, such as those who have hepatic cirrhosis with ascites and metabolic alkalosis or those with resistant oedema. Therefore, careful monitoring of serum electrolytes and BUN levels is important when Moduride is given to such patients.
Azotaemia: As azotaemia may be precipitated or increased by hydrochlorothiazide, special caution is necessary in patients with impaired renal function to avoid cumulative or toxic effects of the components. If increasing azotaemia and oliguria occur during treatment Moduride should be discontinued.
Hepatic disease: In patients with pre-existing severe liver disease hepatic encephalopathy, manifested by tremors, confusion, and coma, and increased jaundice have been reported in association with diuretic therapy including amiloride HCl and hydrochlorothiazide.
Metabolic: Hyperuricaemia may occur or gout may be precipitated in certain patients receiving thiazide therapy. Because calcium excretion is decreased by thiazides, Moduride should be discontinued

before carrying out tests for parathyroid function. Pathological changes in the parathyroid glands with hypercalcaemia and hypophosphataemia have been observed in a few patients on prolonged thiazide therapy; however, the common complications of hyperparathyroidism such as renal lithiasis, bone resorption, and peptic ulceration have not been seen. Increases in cholesterol and triglyceride levels may be associated with thiazide diuretic therapy.

Sensitivity reactions: Sensitivity reactions to thiazides may occur in patients with or without a history of allergy or bronchial asthma. The possibility of exacerbation or activation of systemic lupus erythematosus has been reported with the use of thiazides.

Non-melanoma skin cancer: An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide (HCTZ) exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry. Photosensitizing actions of HCTZ could act as a possible mechanism for NMSC.

Patients taking HCTZ should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions. Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies. The use of HCTZ may also need to be reconsidered in patients who have experienced previous NMSC.

Acute Respiratory Toxicity: Very rare severe cases of acute respiratory toxicity, including acute respiratory distress syndrome (ARDS) have been reported after taking hydrochlorothiazide. Pulmonary oedema typically develops within minutes to hours after hydrochlorothiazide intake. At the onset, symptoms include dyspnoea, fever, pulmonary deterioration and hypotension. If diagnosis of ARDS is suspected, Moduride should be withdrawn and appropriate treatment given. Hydrochlorothiazide should not be administered to patients who previously experienced ARDS following hydrochlorothiazide intake.

INTERACTION WITH OTHER MEDICATIONS: Angiotensin-Converting Enzyme Inhibitors/Amiloride HCl: When amiloride HCl is administered concomitantly with an angiotensin-converting enzyme inhibitor, cyclosporine or tacrolimus, the risk of hyperkalaemia may be increased. Therefore, if concomitant use of these agents is indicated because of demonstrated hypokalaemia, they should be used with caution and with frequent monitoring of serum potassium.

Thiazide diuretic – when given concurrently the following drugs may interact with thiazide diuretics.

Alcohol, barbiturates, or narcotics – potentiation of orthostatic hypotension may occur.

Antidiabetic drugs – (oral agents and insulin) – dosage adjustment of the antidiabetic drug may be required.

Other antihypertensive drugs – additive effect. Diuretic therapy should be discontinued for 2-3 days prior to initiation of therapy with an ACE inhibitor to reduce the likelihood of first dose hypotension.

Corticosteroids, ACTH – intensified electrolyte depletion, particularly hypokalaemia.

Pressor amines (eg. noradrenaline) – possible decreased response to pressor amines but not sufficient to preclude their use.

Skeletal muscle relaxants, nondepolarising (eg. tubocurarine) – possible increased responsiveness to the muscle relaxant.

Lithium – generally should not be given with diuretics. Diuretic agents reduce the renal clearance of lithium and add a high risk of lithium toxicity. Refer to the package inserts for lithium preparations before use of such preparations.

Nonsteroidal Anti-Inflammatory Drugs – In some patients, the administration of a nonsteroidal anti-inflammatory drugs agent can reduce the diuretic, natriuretic and antihypertensive effects of diuretic. Concomitant administration of non-steroidal anti-inflammatory drugs (NSAIDs) and potassium-sparing agents, including amiloride HCl may cause hyperkalemia and renal failure, particularly in elderly patients. Therefore, when amiloride HCl is used concomitantly with NSAIDs, renal function and serum potassium levels should be carefully monitored.

Cholestyramine and colestipol resins – absorption of hydrochlorothiazide is impaired in the presence of anionic exchange resins. Single dose of either cholestyramine or colestipol resins bind the hydrochlorothiazide and reduce its absorption from the gastrointestinal tract by up to 85 and 43 percent, respectively.

USE IN PREGNANCY & LACTATION: Because clinical experience is limited, Moduride is not recommended for use during pregnancy. Thiazides appear in breast milk. If use of the drug is deemed essential, the patient should stop nursing. Thiazides cross the placental barrier and appear in the cord blood. Therefore, the use of Moduride when pregnancy is present or suspected requires that the potential benefits of the drug must be weighed against possible hazards to the foetus. These hazards include foetal or neonatal-jaundice, thrombocytopenia and possibly other side effects that have occurred in the adult. The routine use of diuretics in otherwise healthy pregnant women with or without mild oedema is not indicated.

SIDE EFFECTS: Body as a whole – headache, weakness, fatigue, malaise, chest pain, back pain, syncope.
Cardiovascular – arrhythmia, tachycardia, digitalis toxicity, orthostatic hypotension, angina pectoris.
Digestive – nausea / anorexia, vomiting, diarrhoea, constipation, abdominal pain, GI bleeding, appetite changes, abdominal fullness, flatulence, thirst, hiccups.
Metabolic – elevated serum potassium levels (>5.5 mmol / L), electrolyte imbalance, hyponatraemia, gout, dehydration, symptomatic hyponatraemia.
Integumentary – rash, pruritus, flushing, diaphoresis.
Musculoskeletal – leg ache, muscle cramps, joint pain.
Nervous – dizziness, vertigo, paraesthesia, stupor.
Psychiatric – insomnia, nervousness, mental confusion, depression, sleepiness.
Respiratory – dyspnoea.
Special senses – bad taste, visual disturbance, nasal congestion.
Urogenital – impotence, dysuria, nocturia, incontinence, renal dysfunction including renal failure.
Eye disorders – frequency 'not known': Choroidal effusion, acute myopia, acute angle-closure glaucoma
Respiratory, thoracic and mediastinal disorders – Frequency 'very rare': Acute respiratory distress syndrome (ARDS)

Neoplasms benign, malignant and unspecified (incl cysts and polyps)
Frequency 'not known': Non-melanoma skin cancer (Basal cell carcinoma and Squamous cell carcinoma)

Description of selected adverse reactions
Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed.

SYMPTOMS AND TREATMENT OF OVERDOSE: Amiloride HCl: The most common signs and symptoms to be expected with overdosage are dehydration and electrolyte imbalance. If hyperkalaemia occurs, active measures should be taken to reduce the serum potassium levels.

Hydrochlorothiazide: The most common signs and symptoms observed are those caused by electrolyte depletion (hypokalaemia, hypochloraemia, hyponatraemia) and dehydration resulting from excessive diuresis. If digitalis has been administered, hypokalaemia may accentuate cardiac arrhythmias.

No specific information is available on the treatment of overdosage with Moduride, and no specific antidote is available. Treatment is symptomatic and supportive. Therapy with Moduride should be discontinued and the patient observed closely. Suggested measures include the induction of emesis and / or gastric lavage.

STORAGE CONDITIONS: Store below 30ºC. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE: 10 X 10's in blister pack.

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA